



Zamson Lim Zhen Pin

FINAL-YEAR CYBERSECURITY STUDENT · SOC & DETECTION ENGINEERING

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Final-year cybersecurity student at Asia Pacific University. I joined an employer that had been hit by ransomware months earlier with no monitoring in place, proposed a Security Operations Centre, got approval, and **built it as the only person on security** in four months. **Malware development for purple-team testing is the part I care most about** — writing the payload and detonating it in an isolated lab tells you exactly what your detections see and what they miss.

Looking for: a graduate SOC analyst or detection engineering role.

EXPERIENCE

Network & Security Administrator (Intern)

May 2026 – Aug 2026

Palmgold Management Sdn Bhd · Petaling Jaya · sole owner of security operations

BUILT

- Took the SOC from a proof of concept on a spare desktop to production, and handed it over documented and running at the end of the internship.
- Deployed a Wazuh SIEM with co-located OpenSearch on a Proxmox host I set up myself, integrating firewall, core and access switching, wireless, Windows endpoint, Active Directory and Zeek telemetry through custom ingest pipelines.
- Rolled out Wazuh agents across 80+ domain-joined Windows machines through Group Policy, with an idempotent installer that was safe to re-run and logged per machine.
- Built the network sensor: SPAN feeds from the core and access switches into a multi-node Zeek cluster through a Python decapsulation stage, so the sensor sees the network instead of a slice of it.

DETECTION & PURPLE TEAM

- Wrote the detection content: custom decoders and correlation rules for the firewall, switches, wireless, Active Directory and Zeek, built on native Windows event logs rather than Sysmon, which I had no authority to push fleet-wide.
- Ran authorized purple-team exercises, attacking from Kali Purple then detecting from the SIEM, and turned each into detection content; surfaced an unmonitored network segment that no rule could have covered.
- Completed an authorized internal vulnerability assessment and delivered three reports, one per audience, downgrading one finding to “unconfirmed” when it would not reproduce.

OPERATIONS

- Recovered a full ingestion outage in which archive logging filled the disk and silently flipped every index read-only, then installed retention policies so it could not recur.
- Restored every agent’s group membership from the daily backup after a Wazuh upgrade silently reset them all.
- Traced recurring link drops and transmit hangs on the SOC host to the network card’s power-saving and offload features, and made the fix survive reboots.

SKILLS

SIEM	Wazuh (deployment, agent management, custom decoders and correlation rules), OpenSearch query DSL and aggregations, ingest pipeline design, file integrity monitoring, alert triage and incident response
NETWORK	Zeek network security monitoring, FortiGate policy and syslog integration, Aruba switching and wireless, traffic mirroring and decapsulation, Wireshark, Cisco Packet Tracer
SYSTEMS	Proxmox virtualization, Ubuntu / Rocky Linux administration, systemd tuning, Nginx reverse proxy and TLS, Windows Server, Active Directory, Group Policy software deployment
SECURITY	Purple-team exercise design and execution, authorized vulnerability assessment (OpenVAS, Nmap, Nikto), malicious code analysis in isolated labs, Autopsy forensics, Kali Linux
CODE	Python, JavaScript (D3.js, Three.js), Bash, Windows batch, Java, C++, C#, Rust
LANGUAGES	English, Mandarin, Malay, Cantonese

BSc (Hons) Computer Science (Cyber Security)

2024 – 2027

Asia Pacific University · Kuala Lumpur · final year

- Coursework across network security, digital forensics and secure development.

ELITE TecForte SOC Programme

Mar 2026

Certificate of completion

Capture The Flag

2024 – present

SekaiCTF · CYNX · BeSec · picoCTF

- SekaiCTF 2025 with team Digital Phantoms: solved a medium reverse-engineering challenge on an Android APK using jadx-gui, keytool and Python.

PROJECTS**DETECTION & PURPLE TEAM****Purple team, both chairs**

- Ran the red side from Kali Purple — a C2 listener behind a tunnelling service, implants beaconing on a fixed interval — then switched sides and hunted it from the SIEM.
- Wrote the samples myself in isolated lab VMs — ransomware behaviour in Rust, a browser credential stealer in Python, a C2 client in C++ — and turned each into Wazuh rules, Zeek scripts and OpenSearch queries.
- Firewall and tunnel logs showed only shared addresses; DNS was the only source that named the infected machines. A plain implant over HTTPS to a clean cloud server was still **missed**.

Authorized vulnerability assessment

- Internal assessment from Kali Purple using OpenVAS, with Nmap, Nikto and command-line work to verify by hand what the scanner claimed.
- Scope was the office network, including printers and copiers; the specific findings stay with the company.

COLLECTION & INFRASTRUCTURE**Zeek network monitoring pipeline**

- Rewrote the Zeek decoders to classify each log stream by the fields unique to it, with readable alert descriptions.
- Detections for domain-generation bursts, DNS tunnelling, exfiltration and inbound scans.
- Showed that an alarming reported capture loss was an MTU artefact of the mirror tunnel, not missing traffic, so connection and DNS logs could still be trusted.

Alerting pipeline

- Custom Python integration pushing critical alerts to chat, so triage never depended on somebody watching a dashboard.
- Alerts are de-duplicated with file locking, each send is checked against the chat API response, and binary content no longer breaks it.

TOOLING & FORENSICS**SOC operator dashboards**

- Three single-file consoles built for daily use: a D3 world traffic map, an endpoint alert console, and a Three.js traffic visualizer.
- De-identified and published as open-source templates; live demos and source linked from the portfolio.

Ransomware canary detection

- File integrity monitoring rules scoped to a canary folder, mapped to MITRE T1486, to catch mass-encryption behaviour early.
- Validated with a benign encryptor hard-guarded to that folder — never a live sample.
- The first version fired on every agent's overnight scheduled scan across the fleet; scoping it to the canary folder fixed that.

Active Directory monitoring

- Learned AD properly — domain structure, Group Policy, event log semantics — then built monitoring on native Windows event logs.
- Saved searches for authentication failure, log clearing and account enumeration, ranked by volume so an unusual spike surfaces early.

Wireless & access point monitoring

- Wireless syslog brought into the SIEM with custom decoders for association, authentication and roaming events.
- A scheduled enrichment pipeline resolving hardware addresses to hostnames, so an alert names a machine rather than a MAC.

Malicious file analysis (Autopsy)

- Forensics exercise at APU examining a lecturer-provided sample in Autopsy.
- Worked through the file's contents and metadata rather than trusting a scanner verdict; written up as a formal report.